Joshua T. Paddock

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TECHNICAL SKILLS

- **Embedded & Edge Hardware:** Jetson Orin Nano Super, Raspberry Pi Zero 2W, ESP32
- **Programming:** Python, C++, C, MATLAB, ROS2

EXPERIENCE

Ontario Power Generation *April 2025 – August 2026*
Fuel Handling Engineering Student *Bowmanville, ON*

- Engineered a Python-based validation tool to map command density fields for X, Y, and rotational motion, transforming raw machine logs into heatmaps that defined optimal bench-testing conditions for new inverter-driven motors across 8 fueling machines, with results signed off by senior engineering.
- Redesigned a heavy-water (D_2O) drip tray to protect the Z-axis motors of the fueling machine, using rapid 3D prototyping to engineer an adaptive mounting configuration and a custom gamma-resistant cable guard rated for a 40-year life, validated for fit on 8 machines with zero impact on travel range.
- Deployed a quantized Llama 3.1 8B model for CPU-only local inference, keeping proprietary data on-premises, and implemented fuzzy-matched keyword extraction to minimize compute utilization.
- Created FH team's first single source of truth for failure-screening corrective actions, automating an engineer-validated Power Automate tracker that ended management follow-ups and cut meetings 50%.
- Led field data collection for a University of Waterloo/UNENE trolley-coasting study, groundwork for a planned $5.6M predictive maintenance initiative, and reconstructed trolley position from raw Gray-code encoder signals in Python for the research team's analysis.
- Acted as sole engineering field MTL (Modification Team Leader) for one of three execution windows in a $19M relay panel overhaul (∼600 relays), coordinating with remote engineering teams to troubleshoot schematics in real time, resolving delays that saved 96 hours of project time.

Motion Pac International *May – August 2024*
Mechanical Engineer Intern *Cobourg, ON*

- Designed and optimized machinery for 12+ projects in SolidWorks, applying GD&T for manufacturability.
- Led automated solution for weld and leak-test system with projected 8-month ROI and improved efficiency.

Spoolon Manufacturing Limited *June – August 2022 & 2023*
Mechanical Engineer Intern *Cobourg, ON*

- Produced 250+ AutoCAD mechanical drawings, digitizing legacy factory stamps and dies.
- Optimized spool designs with relief holes in SolidWorks FEA, reducing material waste by 5%.

PROJECTS

Capstone: Adaptive Routing for Language Model Inference Clusters *September 2026 – Present*
Team of 5 | Supervisors: Prof. Serdar Yüksel, Prof. Ian Karlin

- Formulating request routing for a mixture-of-experts language model cluster as a stochastic control problem (MDP), where a policy predicts each request's compute load and routes it to servers with spare capacity, maximizing throughput under latency targets.

Edge AI & Robotics Platform *June 2026 – Present*
Jetson Orin Nano Super, SO-ARM101, Python, 3D Printing

- Deployed and benchmarked local LLM inference on a Jetson Orin Nano Super, evaluating quantized model performance under edge memory and thermal constraints.
- Built an open-source SO-ARM101 leader-follower arm + wrist and overhead cameras to train ACT policies from teleoperated demos; extending Jetson into real-time controller for the follower arm.

EDUCATION

Queen's University, Smith Faculty of Engineering *September 2022 – May 2027*
B.A.Sc. in Mathematics & Mechanical Engineering *Kingston, ON*

- *Coursework:* Control Theory, Intro to Robotics, Optimization & Control, Optimization Theory ML